

Interim Reviewer-Response Evidence and Publication Readiness Audit

Adaptive Robust Time-Series Ensembles: Synthetic + Safi Scope

Evidence compiled from the current Julia publication run

30 July 2026

Abstract

This report evaluates whether the currently successful experiment artifacts are sufficient to answer the mixed NeurIPS reviewer feedback. The answer is **not yet for every question**. The run has completed the expanded 30-seed synthetic regime study, the corrected manuscript sweeps, the Julia-native meta-learning comparison, the Safi application, Safi sensitivity, and all 420 stress-test tasks. The stress stage failed only while converting a numeric setting label to **String**; no stress model failed. Same-machine runtime benchmarking, controlled solver verification, and final artifact verification have not yet run. The current evidence therefore resolves most questions about drift, baselines, sensitivity, limited data, correlated members, and cold starts, but it does not yet resolve the reviewers' computational-cost request and does not support renewed energy or hurricane claims.

Contents

1	Executive assessment	2
1.1	Bottom line	2
1.2	Run status and provenance	2
1.3	Reviewer-question coverage matrix	2
2	Expanded synthetic evidence	4
2.1	Nine drift regimes	4
2.2	Scientific interpretation	5
2.3	How this answers the reviewer	5
3	Modern baseline and meta-learning comparison	6
3.1	Completed comparison	6
3.2	Findings	6
4	Safi real-world evidence	8
4.1	Held-out results	8
4.2	Quantitative conclusions	8
5	Hyperparameter sensitivity	10
5.1	Safi $\tau \times \lambda$ surface	10
5.2	Synthetic window sweep	11
6	Limited-history behavior	12
7	Correlated members and pruning	13
8	Cold start and event boundaries	14
9	Baseline selection and fairness	16
10	What remains before a complete reviewer response	17
10.1	Runtime and computational burden	17
10.2	Exact numerical verification	17
10.3	Empirical scope	17
10.4	Weighted voting	17
11	Recommended revised claims	18
11.1	Claims supported now	18
11.2	Claims that should be removed or moderated	18
12	Pipeline failure and recovery status	19
13	Conclusion	20
A	Artifact inventory used in this report	21

Chapter 1

Executive assessment

1.1 Bottom line

The successful artifacts are enough to produce a strong, candid reviewer response on most empirical concerns. They are not enough to assert that every reviewer request is resolved. In particular:

- The broadened drift study is complete and materially changes the scientific conclusion: Adaptive Ridge is excellent under periodic and recurring reliability patterns, but can fail badly under unprecedented abrupt, persistent-bias, member-degradation, and variance-burst shifts.
- Modern comparators are complete. RLS and Rolling Ridge dominate abrupt shifts; a neural error-history gate narrowly beats Adaptive Ridge in recurring regimes and beats it on Safi.
- Window, regularization, limited-history, correlation, pruning, and event-boundary studies are available.
- The current scope includes one verified real-world application, Safi. Energy was deliberately removed from the active profile after uncertified IRLS and interrupted SOCP solves; hurricane matrices are absent.
- Same-machine runtime comparisons and the final exact solver-verification stage remain pending. These are necessary to answer the computational-burden criticism fully.

1.2 Run status and provenance

The evidence comes from run fingerprint

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The following are complete: 270 expanded-synthetic tasks, 1,020 manuscript-sweep tasks, 120 synthetic-meta tasks, the Safi benchmark, the Safi sensitivity grid, and 420 stress tasks. The stress aggregation failed after all tasks completed because `String::Float64` is undefined for numeric setting labels. The corrected code uses `string(value)` and regenerates the stress summary and figures without recomputing the 420 tasks.

1.3 Reviewer-question coverage matrix

Reviewer concern	Status	Current evidence and qualification
Additional abrupt, periodic, bias, and member-specific drift	Complete	Nine regimes, 30 seeds each, 12 methods. Directly answers the principal synthetic-design criticism.
Modern ensemble and meta-learning baselines	Complete	RLS, Rolling Ridge, Fixed Share, DMA, Hedge, genuine Exp3, neural gate, and boosted meta-learner are included.
Fair tuning protocols	Substantial	Chronological validation grids and selected-hyperparameter tables exist for each method. The paper must print these grids and clearly separate deployable validation selection from the test oracle.

Reviewer concern	Status	Current evidence and qualification
Sensitivity to τ and λ	Complete	Safi has 72 chronological $\tau \times \lambda$ cells; the corrected manuscript sweep evaluates 10 window sizes over 30 seeds.
Limited historical data	Complete	Six history sizes over 30 seeds; results strongly favor RLS or Rolling Ridge below several hundred observations.
Highly correlated members and pruning	Complete	Seven correlation levels, five pruning thresholds, 30 seeds. Numerical conditioning is reported.
Initial history and event boundaries	Complete	Reset, carry, and cross-event borrowing policies are compared over 30 seeds, including first-10-point RMSE.
Computational cost and complete-search runtime	Pending	The runtime stage did not start. Historical paper timings are not a controlled comparison against the new implementation and baselines.
Exact IRLS/JuMP/Gurobi verification	Pending	Controlled verification stage did not start. This is implementation assurance rather than a primary reviewer experiment, but it is required by the repository's final gate.
Breadth across all three historical applications	Pending	Current verified scope is Safi only. Energy and hurricane claims must be removed or explicitly labeled historical and not regenerated.
Weighted-voting extension	Substantial	A softmax/simplex version is conceptually straightforward, but no weighted-voting experiment is in the current artifact.

Chapter 2

Expanded synthetic evidence

2.1 Nine drift regimes

The original synthetic design emphasized gradual drift. The current suite holds the target-generation structure fixed and varies the member-error process across nine regimes. Each regime uses 30 seeds and a chronological train/validation/test protocol.

Table 2.1: Adaptive Ridge versus the best deployable comparator in each expanded regime. Rank excludes the test-oracle member.

Regime	Adaptive RMSE	Best deployable method	Best RMSE	Adaptive rank
stationary	0.0396	RLS	0.0383	4/11
gradual	0.1019	Passive-Aggressive	0.0945	4/11
abrupt	0.7994	RLS	0.0661	10/11
periodic	0.0487	Adaptive Ridge	0.0487	1/11
persistent bias	0.5586	Rolling Ridge	0.0599	9/11
member degradation	0.5874	RLS	0.0823	9/11
recurring	0.0271	Adaptive Ridge	0.0271	1/11
variance burst	0.2511	RLS	0.0784	9/11
discrete gaussian	0.0935	Static Ridge	0.0916	5/11

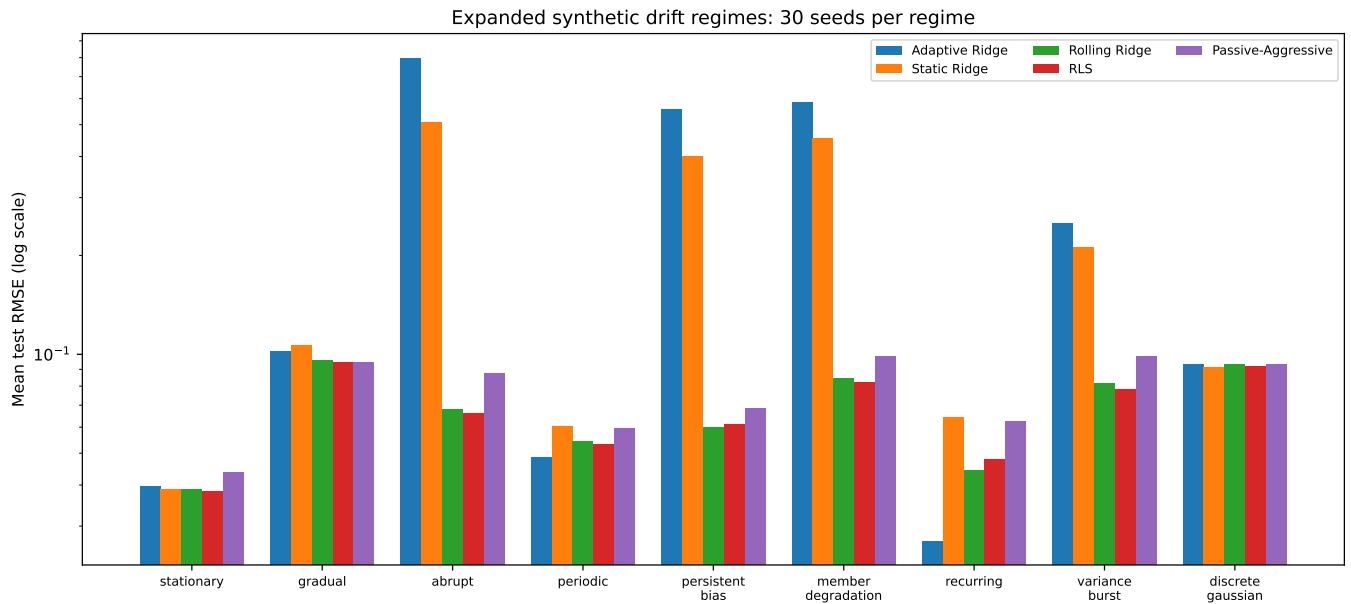


Figure 2.1: Mean test RMSE across the expanded synthetic regimes. A logarithmic vertical axis makes the large abrupt-shift failures visible without obscuring smaller stationary and recurring-regime differences.

2.2 Scientific interpretation

The result is not universal robustness. The defensible claim is conditional:

The affine error-history rule is most effective when recent relative-reliability patterns recur or evolve predictably. Recursive or short-window online methods are safer when the test regime contains an unprecedented discontinuous change.

This statement is supported by paired 30-seed intervals. Under periodic drift, the comparator-minus-Adaptive-Ridge RMSE differences are positive for RLS (0.00464, 95% interval [0.00292, 0.00636]) and Rolling Ridge (0.00576, [0.00395, 0.00757]). Under recurring regimes, Rolling Ridge is worse by 0.01741 ([0.01452, 0.02030]) and RLS by 0.02094 ([0.01728, 0.02460]). Conversely, under abrupt change, RLS is better by approximately 0.733 and wins every seed; similar one-sided failures occur under persistent bias, member degradation, and variance bursts.

2.3 How this answers the reviewer

The expanded design directly answers the request for sudden regime shifts, periodic drift, persistent bias, and member-specific degradation. More importantly, it discovers failure modes rather than merely adding favorable examples. The paper should replace broad language such as “robust to drift” with a statement about predictable or recurring member-error dynamics.

Chapter 3

Modern baseline and meta-learning comparison

3.1 Completed comparison

The Julia suite includes both lightweight online methods and nonlinear error-history learners. Table 3.1 reports the four regimes selected for the meta-learning comparison.

Table 3.1: Modern/meta comparison over 30 seeds.

Regime	Method	RMSE	CVaR 5%
gradual	RLS	0.0949	0.2232
gradual	Rolling Ridge	0.0961	0.2255
gradual	Adaptive Ridge	0.1019	0.2384
gradual	Neural gate	0.1020	0.2389
gradual	Boosted meta	0.1180	0.2773
abrupt	RLS	0.0661	0.1705
abrupt	Rolling Ridge	0.0682	0.1806
abrupt	Neural gate	0.3514	0.6976
abrupt	Boosted meta	0.3923	0.8419
abrupt	Adaptive Ridge	0.7994	1.6446
member degradation	RLS	0.0823	0.2043
member degradation	Rolling Ridge	0.0848	0.2102
member degradation	Neural gate	0.3344	0.8986
member degradation	Boosted meta	0.3413	0.9473
member degradation	Adaptive Ridge	0.5874	1.7664
recurring	Neural gate	0.0253	0.0658
recurring	Adaptive Ridge	0.0271	0.0726
recurring	Rolling Ridge	0.0445	0.1404
recurring	RLS	0.0480	0.1543
recurring	Boosted meta	0.0635	0.1686

3.2 Findings

- Gradual drift: RLS is best (RMSE 0.0949), followed by Rolling Ridge (0.0961); Adaptive Ridge is 0.1019.
- Abrupt change: RLS (0.0661) and Rolling Ridge (0.0682) are dramatically safer than all learned error-history gates.
- Member degradation: RLS (0.0823) and Rolling Ridge (0.0848) dominate; the affine rule fails to extrapolate the unprecedented reliability collapse.
- Recurring regimes: the neural gate is best (0.0253), Adaptive Ridge is close (0.0271), and both beat RLS and Rolling Ridge.

These results answer the request for modern baselines, but they require a more nuanced significance claim. Interpretability and optimization provenance remain advantages of Adaptive Ridge; universal predictive superiority does not.

Chapter 4

Safi real-world evidence

4.1 Held-out results

The current Safi run uses 2,548 held-out observations and 15 methods. The journal manuscript describes 8,494 one-hour-ahead forecasts from six fixed members and motivates tail risk in pollution management. The new run retains the strictly chronological protocol but finds stronger nonlinear comparators than the original baseline set.

Table 4.1: Selected Safi held-out results. Lower is better.

Method	MAE	RMSE	CVaR 5%	CVaR 15%
Neural gate	0.4790	0.6496	1.7726	1.2867
Boosted meta	0.4855	0.6527	1.7624	1.2966
Adaptive Ridge	0.4955	0.6715	1.8376	1.3409
RLS	0.5050	0.6781	1.8318	1.3490
Static Ridge	0.5213	0.7182	2.0274	1.4181
Ensemble Mean	0.5527	0.7831	2.2716	1.5235
Test-oracle member	0.5068	0.7562	2.2682	1.4623

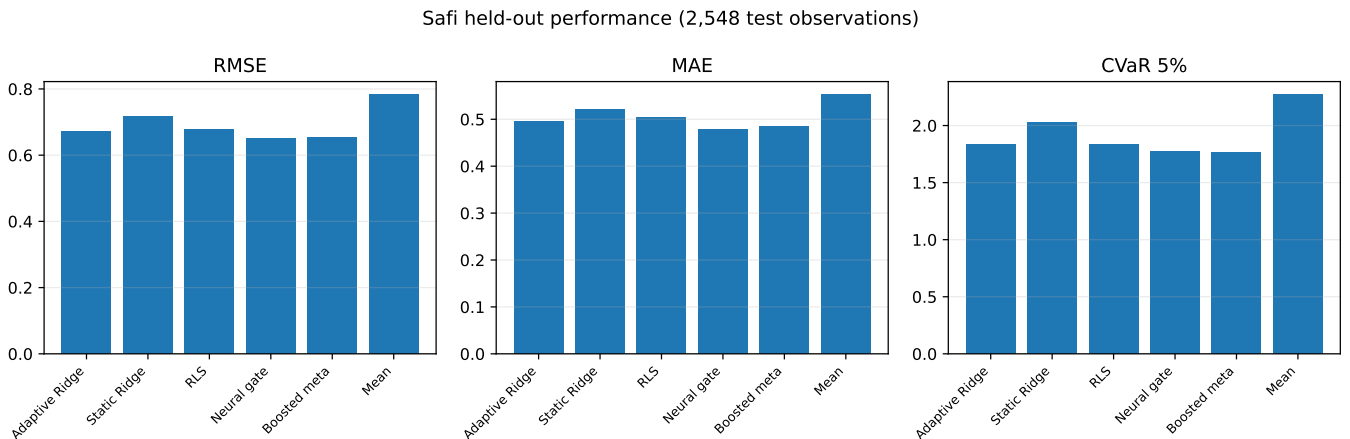


Figure 4.1: Selected Safi methods on average error, squared-error sensitivity, and the worst 5% absolute-error tail.

4.2 Quantitative conclusions

Relative to Static Ridge, Adaptive Ridge reduces RMSE by 6.5%, MAE by 4.9%, CVaR 5% by 9.4%, and CVaR 15% by 5.4%. Relative to the ensemble mean, its RMSE reduction is 14.3% and CVaR 5% reduction is 19.1%. It also improves on the selected individual member by 11.2% in RMSE.

However, the neural gate has RMSE 0.6496 and the boosted meta-learner 0.6527, both below Adaptive Ridge's 0.6715. The 2,000-replication moving-block bootstrap gives a neural-gate-minus-Adaptive-Ridge RMSE interval of

approximately $[-0.0320, -0.0119]$ and a boosted-meta interval of $[-0.0344, -0.0032]$. Thus, the revised paper can claim a clear advantage over static Ridge, equal weighting, and the selected member on this split, but not superiority over modern nonlinear meta-learning.

Chapter 5

Hyperparameter sensitivity

5.1 Safi $\tau \times \lambda$ surface

The Safi sensitivity stage evaluates 9 history windows and 8 regularization levels, for 72 chronological validation cells. The best validation RMSE is 0.35143 at $\tau = 4$ and $\lambda = 0$. Twenty-eight cells are within 1% of the optimum, spanning $\tau = 1$ through 10 and $\lambda = 0$ through 10^{-3} .

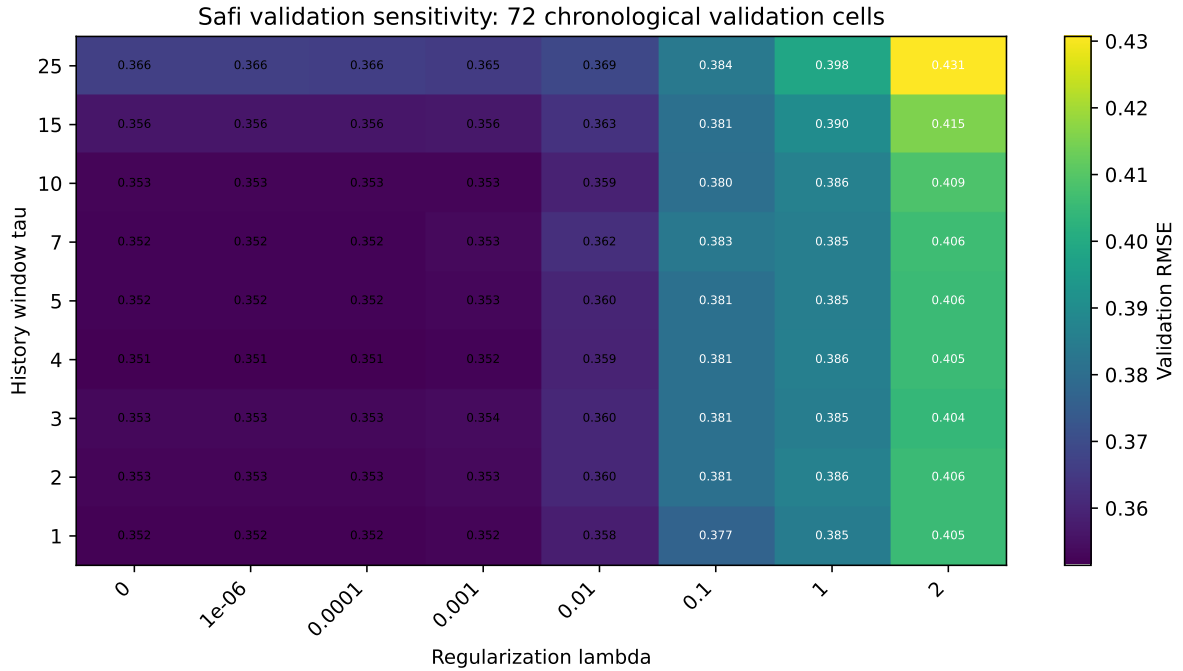


Figure 5.1: Safi validation RMSE. The surface is relatively flat near small regularization and short-to-moderate windows, but degrades at large λ and long windows.

5.2 Synthetic window sweep

Table 5.1: Corrected 30-seed window sweep: mean test RMSE.

τ	Adaptive Ridge	Static Ridge	RLS	Passive-Aggressive
1	0.1018	0.1064	0.0947	0.0974
2	0.1035	0.1064	0.0947	0.0974
3	0.1056	0.1064	0.0947	0.0974
5	0.1100	0.1064	0.0947	0.0974
10	0.1195	0.1064	0.0947	0.0974
15	0.1292	0.1064	0.0947	0.0974
20	0.1373	0.1064	0.0947	0.0974
25	0.1444	0.1064	0.0947	0.0974

Adaptive Ridge RMSE rises from 0.1018 at $\tau = 1$ to 0.1444 at $\tau = 25$. This supports the reviewers' concern about overfitting at large windows and justifies chronological tuning rather than a fixed universal window.

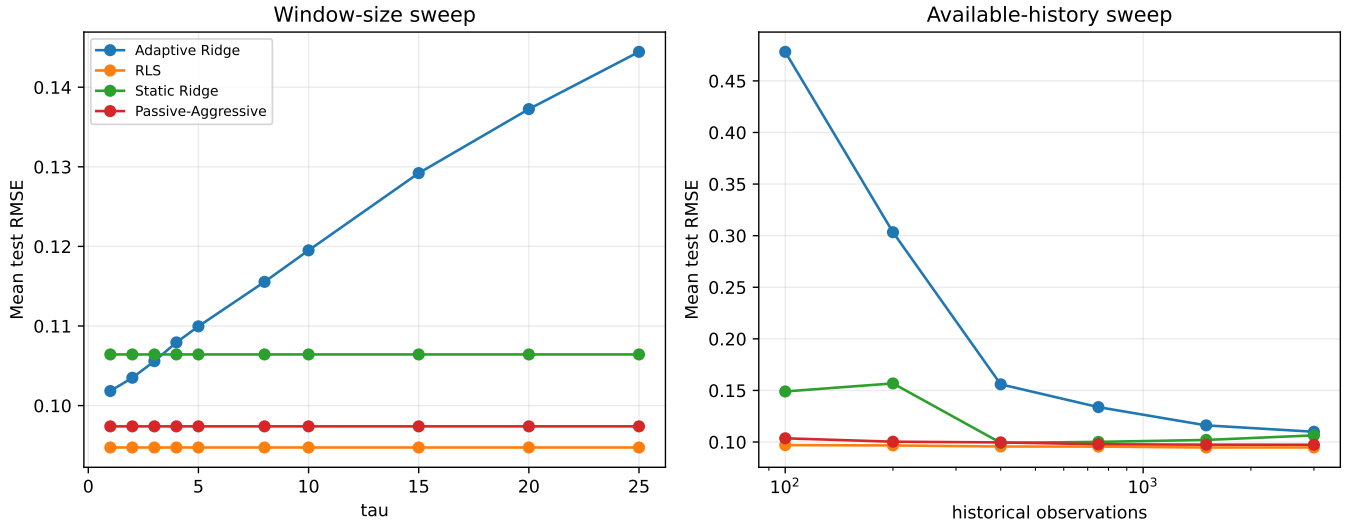


Figure 5.2: Corrected manuscript-style sweeps for window length and available historical observations.

Chapter 6

Limited-history behavior

Table 6.1: Mean test RMSE over 30 seeds as historical data increase.

History	Adaptive	Static	Rolling	RLS	Passive-Aggressive
100	0.3865	0.2860	0.0688	0.0621	0.0770
200	0.5463	0.2647	0.0740	0.0626	0.0781
400	0.0439	0.0731	0.0592	0.0531	0.0701
750	0.0322	0.0683	0.0483	0.0528	0.0674
1500	0.0285	0.0650	0.0449	0.0484	0.0639
3000	0.0271	0.0643	0.0445	0.0480	0.0624

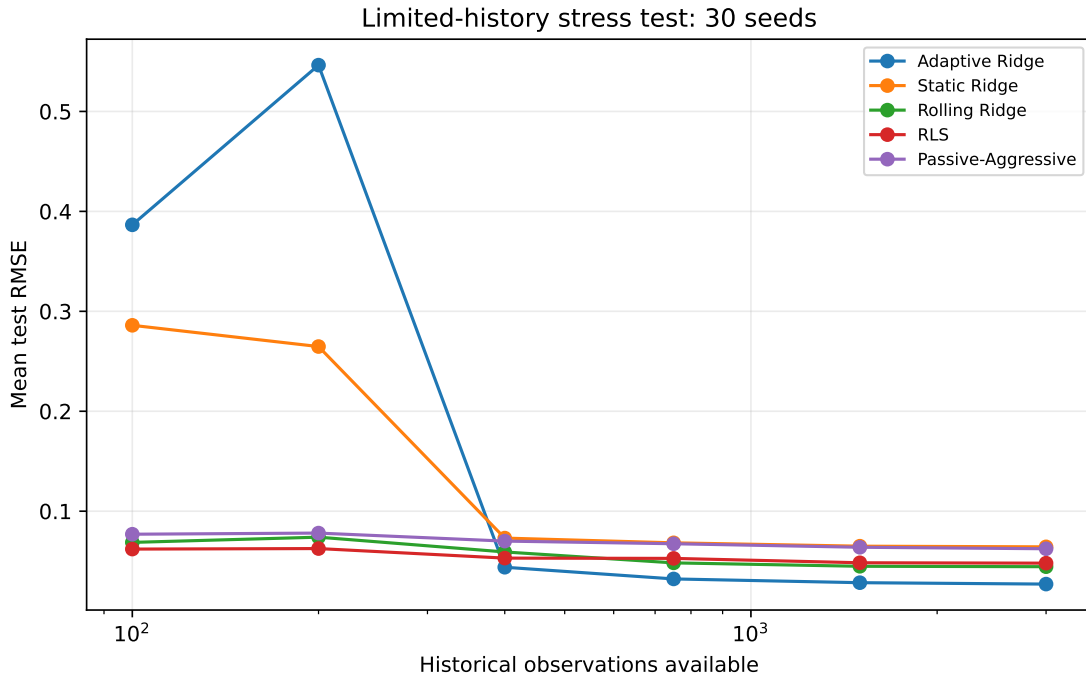


Figure 6.1: Limited-history stress test. The adaptive model is unstable at 100–200 observations, improves sharply by 400, and becomes competitive only with substantially more history.

The current evidence gives a clear answer to the reviewers’ small-data question. The previous broad recommendation that “a few hundred” points suffice is not supported by this corrected experiment. The practical recommendation should be conditional: use RLS, Rolling Ridge, or Static Ridge under limited history unless chronological validation demonstrates a stable adaptive gain.

Chapter 7

Correlated members and pruning

Table 7.1: Correlated-member stress test over 30 seeds. The final column shows that conditioning worsens rapidly even though predictive RMSE remains stable or improves in this generator.

ρ	Adaptive	Adaptive + pruning	Static	RLS	Median condition number
0	0.0271	0.0271	0.0643	0.0480	8.35×10^5
0.25	0.0319	0.0318	0.0749	0.0564	1.40×10^6
0.5	0.0305	0.0305	0.0739	0.0555	2.25×10^6
0.75	0.0253	0.0253	0.0612	0.0502	7.87×10^6
0.9	0.0193	0.0194	0.0428	0.0372	5.76×10^7
0.95	0.0156	0.0155	0.0314	0.0283	2.31×10^8
0.99	0.0093	0.0094	0.0145	0.0134	5.73×10^9

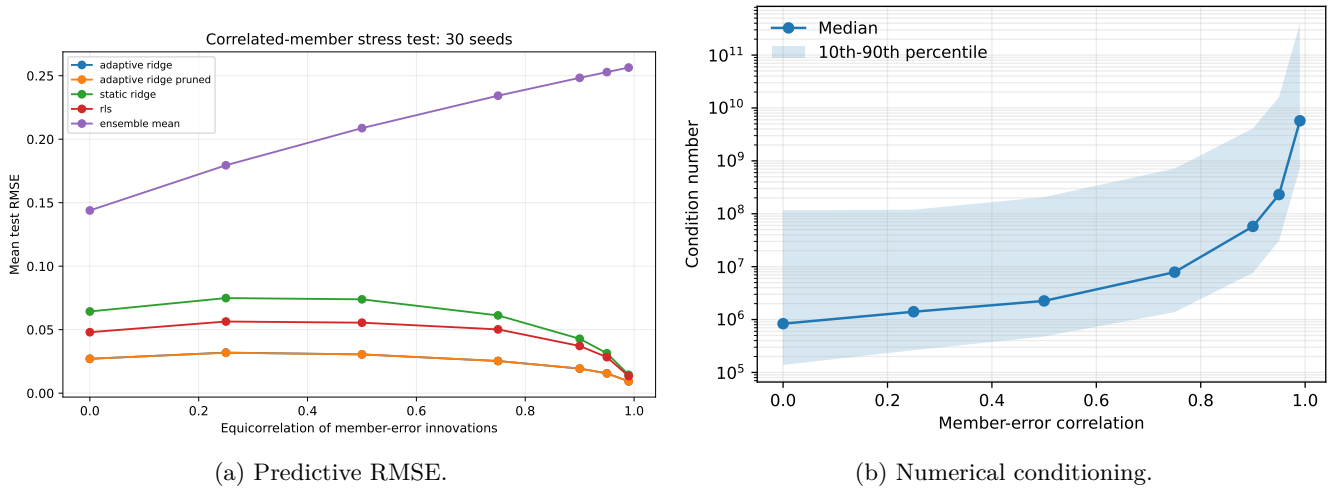


Figure 7.1: Correlation stress evidence.

The main result is numerical rather than a universal predictive penalty: the median condition number increases from roughly 8.4×10^5 at $\rho = 0$ to 5.7×10^9 at $\rho = 0.99$, while the selected regularization and shorter window keep predictions stable. Validation-controlled pruning changes mean RMSE only marginally and usually retains nearly all members. Therefore, automatic aggressive pruning is not supported; conditioning diagnostics and validation-controlled selection are preferable.

Chapter 8

Cold start and event boundaries

Table 8.1: Boundary-policy results over 30 seeds. Cold-start RMSE is computed on the first ten points after event boundaries.

Policy	Method	Overall RMSE	Cold-start RMSE
borrow across events	Adaptive Ridge Borrow Across Events	0.0271	0.0474
carry	Rls Carry	0.0480	0.1383
carry	Passive Aggressive Carry	0.0625	0.0755
carry	Hedge Carry	0.1098	0.1393
reset	Adaptive Ridge Reset	0.0267	0.0382
reset	Rls Reset	0.0502	0.1660
reset	Passive Aggressive Reset	0.0629	0.0890
reset	Hedge Reset	0.1108	0.1633

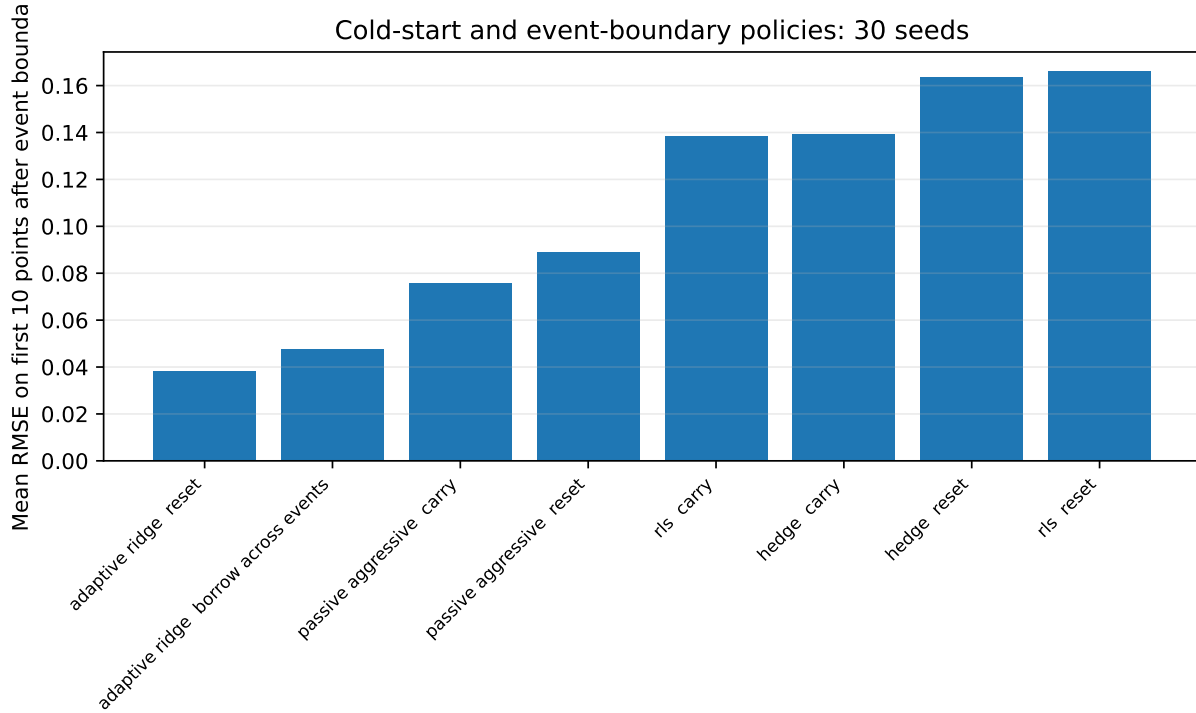


Figure 8.1: Cold-start error under reset, carry, and cross-event borrowing policies.

The adaptive static fallback with event reset has the best cold-start RMSE (0.0382), compared with 0.0474 when histories are borrowed across events. This answers the reviewer’s question about initial validation/test points: the

implementation now uses an explicit fallback until sufficient same-event verified history exists, and the policy is empirically evaluated rather than left implicit.

Chapter 9

Baseline selection and fairness

The current implementation addresses the concern that some dynamic baselines appeared implausibly poor. It distinguishes:

- Hedge, which receives full information about every member's loss;
- genuine bandit-feedback Exp3;
- Fixed Share for tracking a changing best expert;
- RLS and Rolling Ridge as strong lightweight adaptive regressors;
- Passive-Aggressive with chronological margin tuning;
- DMA with tuned forgetting factors;
- neural and boosted error-history meta-learners.

Every tunable method uses the same chronological validation segment and the test set is evaluated once. The test-oracle member is explicitly labeled nondeployable; a validation-selected member is reported separately. This is a substantial answer to tuning fairness, but the revised manuscript should print the exact grids, validation metric, and update-information timing for every method.

Chapter 10

What remains before a complete reviewer response

10.1 Runtime and computational burden

The current artifact is not enough to answer the runtime criticism. The runtime stage was scheduled after stress aggregation and never began. The historical paper’s timing tables cannot be used as controlled comparisons because they mix an older JuMP formulation, different hardware, and no same-machine comparator timings. The corrected run must still produce:

- selected-model fit time;
- complete $\tau \times \lambda$ search time;
- prediction/update latency;
- scaling with n , m , and τ ;
- parameter and memory counts;
- same-machine comparisons against RLS, Rolling Ridge, Fixed Share, Passive-Aggressive, and static Ridge.

10.2 Exact numerical verification

The controlled solver-verification stage also remains pending. It is not required to interpret the completed predictive results, but it is necessary for the repository’s claim that reduced, iterative, and exact JuMP/Gurobi formulations agree within tolerance.

10.3 Empirical scope

The active verified scope is synthetic + Safi. The current NeurIPS draft still describes energy and tropical-cyclone applications, while the present artifact does not regenerate them. The paper must either restore and verify those applications or narrow the abstract, introduction, contribution list, result tables, conclusion, and checklist to the current scope.

10.4 Weighted voting

The reviewer’s weighted-voting question can be answered conceptually: soft voting can use affine scores followed by a simplex projection or softmax, while hard voting leads to a classification objective and does not inherit the current linear-regression robustification equivalence directly. No current experiment validates that extension, so it should be presented as future work rather than a completed contribution.

Chapter 11

Recommended revised claims

11.1 Claims supported now

- Adaptive Ridge is interpretable and materially improves on Static Ridge, equal weighting, and a selected individual member on the Safi split.
- The method is especially effective under periodic and recurring member-reliability patterns.
- Chronological tuning of τ and λ is necessary; long windows and limited history can be harmful.
- Modern recursive and nonlinear baselines define important safety and performance frontiers.
- Explicit cold-start and event-boundary handling removes an implementation ambiguity identified by reviewers.

11.2 Claims that should be removed or moderated

- Remove universal claims of robustness to arbitrary drift.
- Do not claim overall superiority to modern meta-learning; the neural gate and boosted meta-learner outperform Adaptive Ridge on Safi.
- Replace the “few hundred observations” recommendation with a validation-dependent rule; corrected experiments show severe instability at 100–200 observations.
- Do not present energy or hurricane results as regenerated by the current verified package.
- Do not claim the computational-cost concern is resolved until the runtime stage passes.

Chapter 12

Pipeline failure and recovery status

The current run did not fail in any of the 420 stress tasks. It failed after writing the 3,180-row stress table and the 1,050-row pruning table, while constructing a summary label:

```
setting=[String(subset.setting[1])]
```

CSV type inference represented numeric settings as `Float64`. In Julia, `String(x)` is a type constructor/conversion and has no method for `Float64`; `string(x)` is the generic textual representation. The corrected code uses `string` consistently in aggregation, plotting, runtime labels, and artifact verification, and adds a mixed numeric/text regression test.

A guarded migration utility preserves the eight completed stages and all 420 fingerprint-bound stress task outputs. It does not copy old failure logs, the incomplete stress stage marker, wrapper receipts, or serialized model objects. The resumed run performs only stress aggregation/figures, solver verification, runtime benchmarking, and final verification.

Chapter 13

Conclusion

The current result set answers most reviewer questions about empirical robustness, modern baselines, tuning, failure modes, limited data, correlated members, and boundary handling. It is not yet sufficient to answer every question because the controlled runtime study and exact solver-verification stage remain incomplete, and the current empirical scope is Safi rather than all three historical applications. A strong reviewer response is already possible, but it must be candid, narrow the claims, and reserve computational-efficiency claims for the completed repaired run.

Appendix A

Artifact inventory used in this report

- `synthetic_regimes_summary.csv`: 108 method-regime rows.
- `synthetic_paired_rmse_intervals.csv`: 99 paired comparisons.
- `manuscript_sweeps_summary.csv`: 340 sweep-setting-method rows.
- `synthetic_meta_summary.csv`: 20 regime-method rows.
- `real_world_all_metrics.csv`: 15 Safi methods.
- `safi_moving_block_bootstrap.csv`: 2,000-replication paired bootstrap contrasts.
- `sensitivity_safi.csv`: 72 validation cells.
- `stress_tests_per_seed.csv`: 3,180 completed method-task rows.
- `stress_correlation_pruning_validation.csv`: 1,050 threshold candidates.

Bibliography

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